surface acoustic wave sensor

Design and Fabrication of ZnO-Based SAW Sensor Using Low Power Homo-Buffer Layer for Enhanced Humidity Sensing

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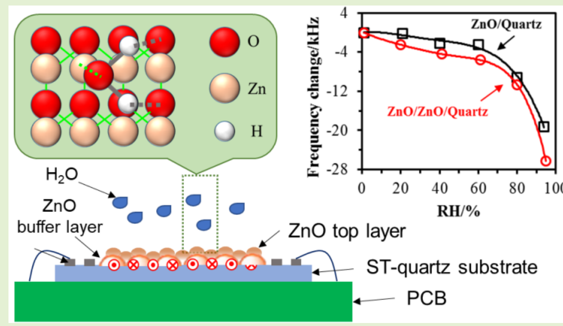
***Abstract*—Zinc oxide (ZnO) nanostructures were proven to enhance the sensitivity performance of nanoscale ZnO-based surface acoustic wave (SAW) sensor greatly. Here, we developed ZnO sensitive layer-based SAW sensors by low power homo-buffer layer (LPHBL) method and focused on the quality of preparation, their mass sensitivity and humidity sensing applications. ZnO films were prepared on the surface of the SAW devices by radio frequency magnetron sputte ring method by employing different sputtering power. Island-shape ZnO nanostructures were successfully prepared by the simple LPHBL method. The mass sensitivities of these sensor devices based on Rayleigh wave mode and Sezawa wave mode were increased dramatically by increasing ZnO top layer thickness. To evaluate the humidity sensing performance, the humidity testing experiments were conducted for a range of moisture concentrations between 1% and 95%. The ZnO/quartz sensor without LPHBL frequency response magnitudes varied between 0-19.25 kHz and insertion loss (IL) shifts changed between 0-0.815 dB for 1%-95% RH while the ZnO/ZnO/quartz SAW sensor with LPHBL showed 0-26.25 kHz response magnitudes and 0-1.878 dB IL shifts at the same conditions. The Love-SAW sensor with LPHBL exhibited more than 36% greater humidity response sensitivity compared to its no intermediate layer counterpart. In addition, the recovery of the ZnO/ZnO/quartz humidity sensing device could reach 91.5%. Overall, the ZnO-based Love-SAW sensor fabricated by the LPHBL method may hold promise for chemical and biological detection applications.**



***Index Terms*—ZnO film, surface acoustic wave, homo-buffer layer, sensitivity, humidity sensing.**

I. Introduction

With the advent of nanotechnology, the low sensitivity of bulk materials has been overcome eventually by developing high surface to volume ratio nanostructured materials [1]–[3]. Among the nanostructured materials, zinc oxide (ZnO) is an important II-VI semiconductor with remarkable electrical and optical properties and wide-range of applications in electronics, catalyzes, acoustics and sensors [4], [5]. The ZnO nanostructure is non-toxic, environmentally friendly and easily prepared material and has been extensively investigated as chemical sensors [6]–[8]. ZnO nanostructured materials including nanowires [9], nanoparticles [10], nanorods [11], nanofibers [12], nanosheets [13], nanoripples [14] and nanoplates [15] have been frequently applied to surface acoustic wave (SAW) sensors for different gases detection including CO [16], NO_2 [17], H_2S [18], NH_3 [19], CH_4 [20] and CH_2O [21]. Many different methods have been reported for the preparation of ZnO nanostructures, including sol-gel processes [22], chemical vapour deposition (CVD) [23], metal-organic chemical vapour deposition [24], epitaxial growth [25], pulsed laser deposition [26], atomic layer deposition [27] and sputtering [28], [29]. Currently, most of the ZnO films were deposited by the CVD and chemical synthesis [30]–[32]. However, challenges still remain in the chemical techniques as their complicated process and poor reproducibility. The features of ZnO depend on the method used to synthesize this metal oxide that in turn affects the surface morphology and electrical property [33], [34]. Among physical vapour deposition (PVD) techniques, sputtering is one of the best and powerful tools with good reproducibility and compatibility for the large-scale production of uniform thin films and planar device on wide-area substrates [35].

In this work, an effective way to improve the quality of sputtered ZnO thin films is obtained by introducing a low power homo-buffer layer (LPHBL), which is different from the

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Argon ions are accelerated by rf electric field to hit the target made of the material to sputter.

standard radio frequency magnetron sputtering and the conventional direct deposition methods. Firstly, the SAW device was designed and fabricated by traditional micro-fabrication technology. ZnO films were prepared on the surface of the SAW devices by RF magnetron sputtering method by employing different sputtering power. Secondly, the quality of ZnO films and frequency response of the SAW devices were characterized. Finally, the mass sensitivity based on different wave mode and humidity sensing performance of the SAW devices were evaluated.

II. Experimental Details

A. Sensors Fabrication and Measurement

The delay line devices were designed on ST-cut quartz substrate (3158 m/s) with $\lambda = 20\ \mu$m for operation at 157.9 MHz and minimum insertion losses. Finger width, spacing, delay line and aperture is 5 μm, 5 μm, 1500 μm and 1800 μm, respectively. The number of input and output electrode pairs of interdigitated transducers (IDTs) is 100. The aluminum (Al) IDTs with a thickness 150 nm were deposited by e-beam evaporation and patterned using conventional UV-light photolithograph. A frequency spectrum analyzer (Agilent 4396B) was used to obtain the frequency signal by stimulation of a generator. The frequency changes, Δf, can be calculated using PC and a data acquisition system.

B. ZnO Layer Deposition and Characterization

ZnO films were prepared on the delay line by RF magnetron sputtering method, from a 3-inch diameter ceramic ZnO target (4N) by employing different sputtering power, the scheme of ZnO preparation is shown in Table I. Prior to the deposition process, the background vacuum, around 9×10^{-4} Pa, was maintained in the chamber. A mixture of both Ar and O_2 at 18 mL/min flow were subsequently employed for deposition at room temperature. The preparation of ZnO thin film was divided into two stages. The parameters of deposition pressure, sputtering power and sputtering time for low power homo-buffer layer (LPHBL) were fixed at 0.5 Pa, 50 W and 1 h. After a 10-minute interval, the top layer was prepared. The deposition pressure and sputtering time of the top layer were fixed at 0.5 Pa and 1 h. The sputtering power of top layer as a variable parameter was set to 50 W (Device A), 100 W (Device B) and 150 W (Device C), respectively. ZnO top layers were characterized by X-ray diffraction (XRD), scanning electron microscope (SEM) and atomic force microscopy (AFM). Finally, the resonance frequencies of devices B with varying ZnO top layer thicknesses were measured to evaluate the sensitivity performance. The ZnO was deposited layer-by-layer on the devices B and the thickness of ZnO top layers was measured by a step profiler.

C. Humidity Testing Setup

The sensors were placed in a hermetic cylinder space (the inner dimensions of the cylinder are $R^2 \times H \times \pi = 10^2 \times 4 \times \pi$ mm^3) with two through-holes for humidity nitrogen (N_2) gas to pass through for sensing. Only the whole detection surface of each sensor was exposed to humidity. RF coaxial cables were used to connect sensors to signal source and transmission spectrum measurements. Pure water steam was mixed with dry pure (99.999%) N_2 to form standard humidity gas (moisture) using commercial humidity generator. The relative humidity (RH) was varied by adjusting the flow rates of the steam (F_W) to dry N_2 (F_N). Briefly, a different RH value (1%-95%) was generated at room temperature and introduced onto the surface of the sensitive layer. The actual relative humidity output could be read directly from the humidity generator, which was almost consistent with the theoretical RH value. The output standard moisture was introduced into the testing chamber through an accurate mass flow meter. The flow rate of input moisture was controlled at 300 mL/min. The gas path of humidity sensing was formed by using stainless steel pipe (Inner diameter of 3 mm) connection. The schematic diagram of humidity sensing system is shown in Figure 1.

TABLE I
Scheme of ZnO Films Preparation Under Different Power

Sensor	Sputtering power		Vacuum	WG	Time
A	BL: 50 W	TL: 50 W	0.5 Pa	Purity: 4N O_2: Ar=1:1; Flow rate: 18 mL/min	BL: 1 h TL: 1 h
B		TL: 100 W			
C		TL: 150 W			
D	No BL	TL: 100 W			TL: 2 h

BL = buffer layer, TL = top layer, WG = working gas, W = Watt, mL = milliliter.

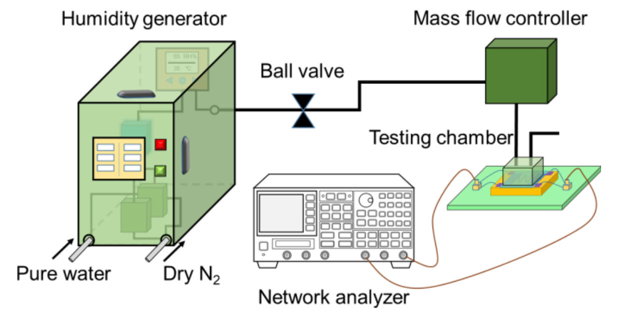

Fig. 1. A schematic of the testing system used for humidity sensing.

III. Results and Discussion

A. Love Wave Sensors Based on ZnO Layers

In current work, the Love wave sensor based on ZnO layers was studied. In order to enhance the preparation quality and preferred orientation of ZnO thin films, the low power homo-buffer layer method was utilized for the preparation of ZnO nanostructures. Figure 2 shows the XRD patterns of ZnO top layers at different condition. Where, curve *a* indicates the XRD pattern of ZnO top layer prepared on the substrate at 100 W without homo-buffer layer for comparison. Curves *b*, *c* and *d* respectively indicates the XRD patterns of ZnO top layers prepared at 50 W, 100 W and 150 W on the low power homo-buffer layer (prepared at 50 W). It can be seen that the ZnO top layer deposited on ST-quartz substrate has several significant diffraction peaks, while the ZnO top layers prepared on the low power homo-buffer layer show a strong (002) diffraction peaks, and other diffraction peaks basically disappear with the increase of sputtering power. Therefore, the

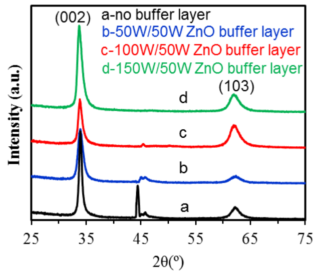

Fig. 2. XRD patterns of ZnO top layers at different condition.

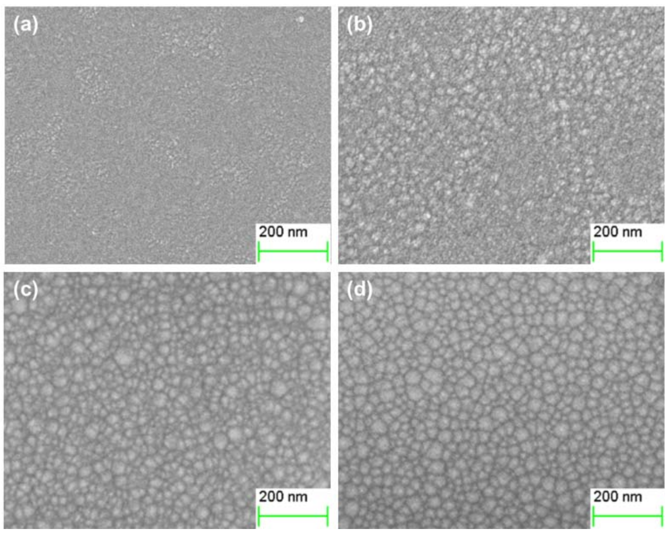

Fig. 3. SEM images of (a) 100 W ZnO/no buffer layer, (b) 50 W ZnO/ZnO (50 W), (c) 100 W ZnO/ZnO (50 W) and (d) 150 W ZnO/ZnO (50 W).

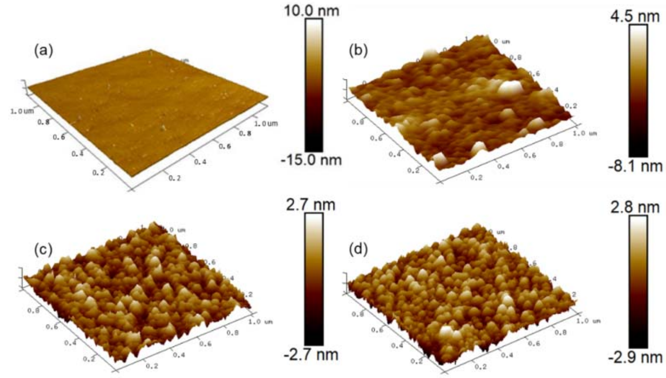

Fig. 4. AFM images of (a) 100 W ZnO/no buffer layer, (b) 50 W ZnO/ZnO (50 W), (c) 100 W ZnO/ZnO (50 W) and (d) 150 W ZnO/ZnO (50 W).

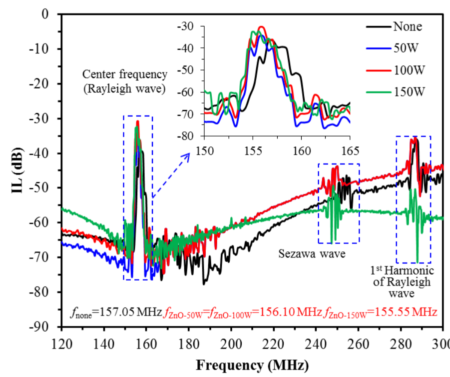

Fig. 5. Transmission spectrum of the SAW devices based on different deposition power.

introduction of low power homo-buffer layer is conducive to the growth of ZnO films in a single crystalline orientation.

Figure 3 shows SEM images selected to show the effect of the sputtering power of the top layer on the morphology of the ZnO/ZnO films. Figures 3(a), (b), (c) and (d) exhibit totally different morphologies with or without homo-buffer layer. The ZnO top layer without homo-buffer layer displays smoother surface with no observable nanostructure (Figure 3(a)). In Figures 3(b), (c) and (d), the island-like structures with nanometer size are seen to uniformly distribute over the whole surface. This evidence partly proves that the homo-buffer layer can greatly improve the degree of crystallinity.

The surface morphology of the ZnO top layers were characterized using atomic force microscopy (AFM). Figure 4(a) shows that the ZnO top layer sputtered on the substrate without homo-buffer layer exhibits flat surface morphology while the ZnO top layers deposited on low power homo-buffer layer sputtered also exhibit obvious island-shape structures, as shown in Figures 4(b) to (d). The main reason is that the growth mechanism of the thin films grown directly on the substrate were different from those of the ZnO top layers grown on the low power homo-buffer layer. When there is no ZnO buffer layer, ZnO nucleates directly on the substrate surface, and its nucleation density is relatively high, which finally leads to the growth of relatively dense thin film structure. When there is a low-power homo-buffer layer, ZnO top layers can be directly epitaxial grown on the buffer layer without nucleation stage, so the grains are easy to grow up and form particles in the deposition process, and the sputtering power has a certain influence on the grain size.

The transmission properties of the SAW devices were characterized and compared with the SAW devices layered by ZnO film at different sputtering power. Figure 5 shows the comparison transmission spectrum of the SAW devices. The deposition time, t, was one hour, and the sputtering power, P, of the top layer was 50 W, 100 W and 150 W, respectively. The resonance frequency, f_r, of the Rayleigh wave was found to be 157.05 MHz for the Rayleigh-SAW device without guiding layer, while the resonance frequency, f_l, of the Love wave was found to be 156.1 MHz, 156.1 MHz and 155.55 MHz for the Love-SAW devices layered by ZnO-50W/ZnO, ZnO(100W)/ZnO and ZnO(150W)/ZnO, respectively. All the devices have the insertion loss (IL) of the resonant peaks over 35 dB, demonstrating their potential for applications in sensors.

B. Mass Sensitivity Evaluation of SAW-Love Mode Devices With Different Thickness of ZnO Top Layer

To evaluate the mass sensitivity performance, the effect of ZnO top layer added layer by layer on the resonance frequency of the Love-SAW device was studied. The sputtering power of the ZnO top layer was fixed at 100 W. The resonance frequencies of Love-SAW devices by varying ZnO top layer thickness were evaluated as shown in Figure 6. The spectra of a series of frequency measurements exhibit two groups of

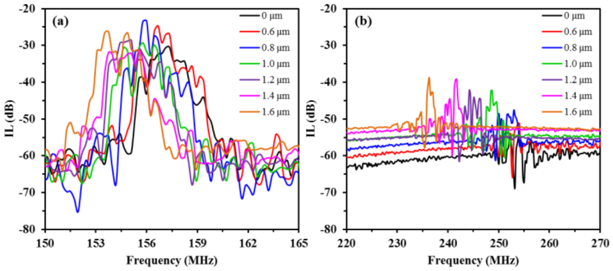

Fig. 6. Influence of different ZnO top layer thickness on frequency of SAW-Love wave sensor: (a) Love wave mode; (b) Sezawa wave mode.

major wave peaks, including Rayleigh wave (or Love wave) and Sezawa wave. It was observed that the Sezawa wave mode has the higher frequency than that of the fundamental Rayleigh mode for the same IDTs. The measurement results were in good agreement with literature reports. The existence of Sezawa wave depends on layered SAW devices based on piezoelectric thin films [36], [37]. The peak of Sezawa wave occurs a little weaker, due to a relative thin ZnO top layer leading to a small difference in SAW phase velocity between SiO_2 substrate and ZnO top layer. The key propagation of SAW should be accumulated in piezoelectric ZnO substrate. As the thickness of deposited ZnO top layers increased especially towards 1.0 μm, the Sezawa wave peak becomes increasingly significant, owing to SAW energy superposition in ZnO top layer and the significant phase velocity difference between substrate and its guiding layer.

The frequencies of both Love wave and Sezawa wave decrease with increasing the thickness of ZnO top layer, as shown in Figure 7. The SAW device generated a resonance frequency of 157.575 MHz with an IL of 30.45 dB, while a Sezawa wave demonstrated a frequency of 254.1 MHz with an IL of 54.48 dB. As the thickness of ZnO top layers increased, both resonance frequency and its corresponding Sezawa wave frequency would shift respectively 2.925 MHz and 17.775 MHz as shown in Figure 7(a) and (b) as ZnO top layer increased from 0 to 1.6 μm. The minimum IL of Love-SAW was obtained with 0.8 μm thickness of a ZnO top layer, while the IL of Sezawa wave decreased with increasing ZnO top layer thickness. However, Sezawa wave mode showed the greater frequency shift than that of Love-SAW mode (Figure 7c) at the same conditions. Mass sensitivity (S_m) was calculated as the frequency shift per Hz per unit area of applied mass. The S_m for SAW devices was given in the report as [38]

$$S_m = \frac{1}{f_c} \lim_{\Delta m \to 0} \frac{\Delta f \cdot A}{\Delta m} = \frac{1}{\rho h} \cdot \frac{\Delta f}{f_c} \tag{1}$$

where f_c is the operating frequency of the SAW device, Δf is the frequency shift, A is the contact area of applied mass, Δm is the applied mass, ρ is the density of the load layer, h is the thickness.

According to the Eq. (1), the mass sensitivity of the Sezawa wave mode increases with the normalized thickness of ZnO top layer, which is apparently higher than that of Rayleigh wave and Love wave mode (Figure 7(d) and (e)). The Rayleigh S_m keeps almost constant values with increasing normalized thickness of ZnO top layers. By contrast, the Sezawa S_m

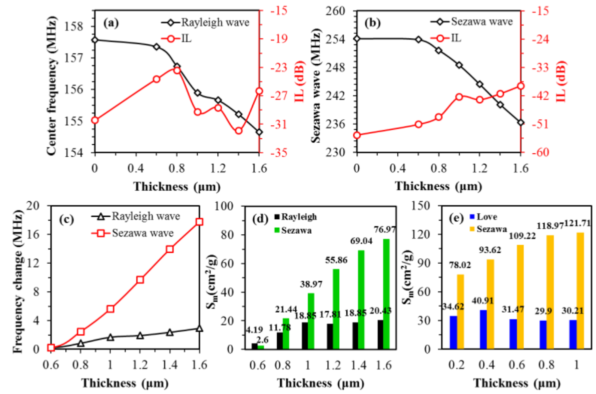

Fig. 7. The frequency change of SAW devices based on varying ZnO top layer thickness: (a) Resonance frequency of SAW-Love device; (b) Sezawa wave mode of SAW-Love device; (c) Comparison of frequency changes between two modes for each ZnO top layer thickness; Mass sensitivity (S_m) of Rayleigh mode (d) and Love mode (e) is compared with the Sezawa mode, respectively.

increases drastically with the same conditions. The average value of S_m of Rayleigh-SAW mode Love-SAW mode and Sezawa wave mode was obtained respectively by 16.82 cm^2/g, 32.1 cm^2/g and 46.33 cm^2/g. The mass sensitivity of Sezawa wave mode shows close to 3 times higher than that of Rayleigh-SAW wave mode, indicating that Sezawa wave mode has also greater application potential in practical detection.

C. Humidity Sensing Performance

To study the sensing performance, the responses of the developed ZnO/quartz and ZnO/ZnO/quartz SAW sensors toward moisture were obtained at a range of RH between 1% and 95% at room temperature. The frequency shift and amplitude change with 6 different RH% (1%, 20%, 40%, 60%, 80% and 95%) can be observed from Figure 8(a) and (b), respectively. It can be seen that the ZnO/quartz sensor without low power homo-buffer layer frequency response magnitudes varied between 0-19.25 kHz and IL shifts changed between 0-0.815 dB for 1%-95% RH while the ZnO/ZnO/quartz SAW sensor with low power homo-buffer layer showed 0-26.25 kHz response magnitudes and 0-1.878 dB IL shifts at the same conditions. Overall, the ZnO/ZnO/quartz SAW sensor showed higher resonance frequency and IL shifts than its ZnO/quartz SAW counterpart for different RH. From Figure 3 or Figure 4, the SAW sensor with low power homo-buffer layer shows the nanostructures-impressed surface compared to the SAW sensor without buffer layer. The rough and wrinkled surface can improve the adsorbability and then the sensitivity of the ZnO/ZnO/quartz sensor.

In Figure 8(c), the frequency response curves of the ZnO/ZnO/quartz humidity sensor were preliminarily recorded on different levels of humidity in the environment using a network analyzer. It follows that the resonance frequency was about 155.48 MHz. The red dashed curve in Figure 8(c) indicates the recovery curve measured under a dry nitrogen atmosphere, which demonstrates the sensor can roughly turn back to its initial state. Figure 8(d) shows the change in resonance frequency as the humidity over the ZnO/ZnO/quartz sensor was switched consecutively between 1% and 80%. The

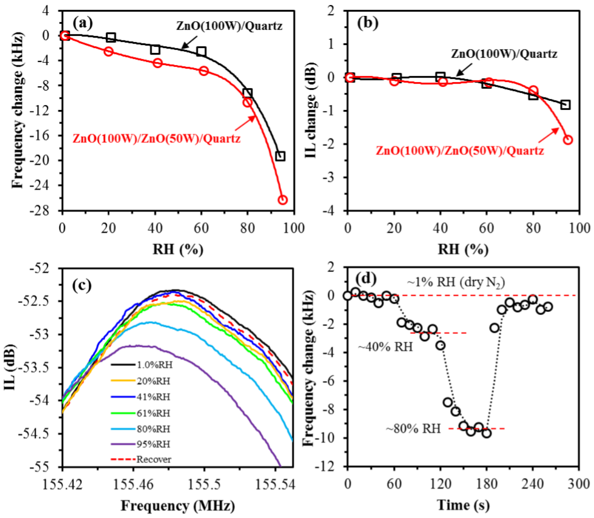

Fig. 8. Response of the ZnO/quartz and ZnO/ZnO/quartz sensors toward relative humidity (1%-95%): (a) resonance frequency and (b) IL changes; (c) the recovery of Love-SAW mode from 95% to 1%; (d) response and recovery times as the humidity over the ZnO/ZnO/quartz sensor was switched between 1%, 40% and 80%.

frequency data was collected every 10 s through the network analyzer, and the change in frequency can then be calculated. Further, the response and recovery times turned out to be 60 s and 40 s respectively. However, multiple measurement results demonstrated that the sensor was unable to completely recover and the best recovery of the sensor was approximately 91.5%.

IV. Conclusion

The fabrication and detection of a ZnO-based humidity sensing device based on an experiment study through an effective low power homo-buffer layer to enhance the sensing performance were presented. ZnO films were prepared on the surface of the SAW devices by radio frequency magnetron sputtering method by employing different sputtering power. Island-shape ZnO nanostructures were successfully prepared by a simple homo-buffer layer method. To evaluate the mass sensing performance, the effect of ZnO top layer added layer by layer on the frequency response of the Love-SAW sensor based on different wave mode was studied. The mass sensitivities of these sensor devices based on Rayleigh wave mode and Sezawa wave mode were increased dramatically by increasing the thickness of ZnO top layer. For humidity application, the Love-SAW sensor with LPHBL exhibited more than 36% greater humidity response sensitivity compared to the conventional SAW device. Further, this sensor also shows good repeatability. Therefore, the Love-SAW sensor fabricated by the LPHBL method may hold promise for chemical and biological detection applications.

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Surface acoustic wave sensors are a class of microelectromechanical systems (MEMS) which rely on the modulation of surface acoustic waves to sense a physical phenomenon. The sensor transduces an input electrical signal into a mechanical wave which, unlike an electrical signal, can be easily influenced by physical phenomena. The device then transduces this wave back into an electrical signal. Changes in amplitude, phase, frequency, or time-delay between the input and output electrical signals can be used to measure the presence of the desired phenomenon.

Surface acoustic wave humidity sensors require a thermoelectric cooler in addition to a surface acoustic wave device. The thermoelectric cooler is placed below the surface acoustic wave device. Both are housed in a cavity with an inlet and outlet for gases. By cooling the device, water vapor will tend to condense on the surface of the device, causing a mass-loading.